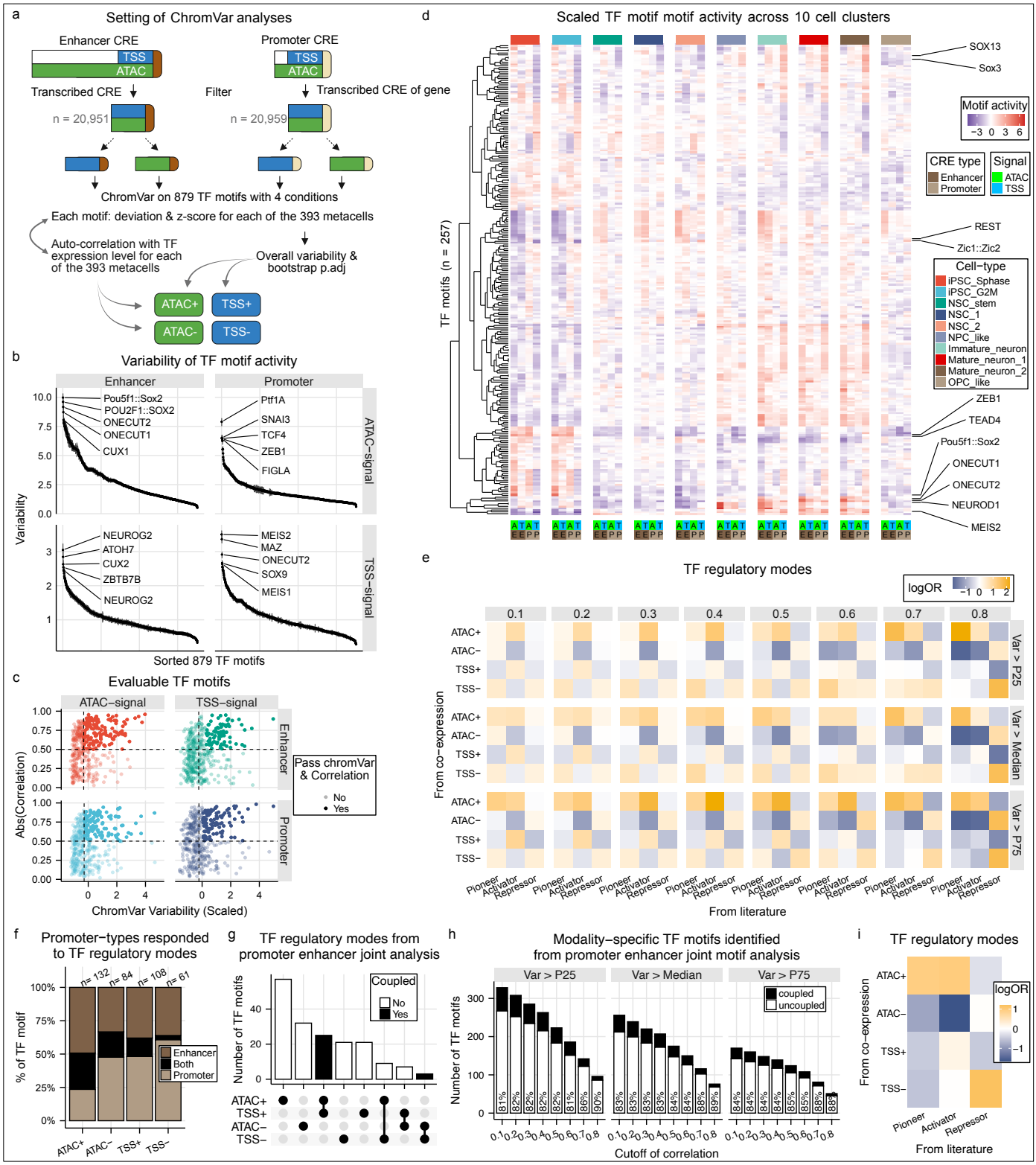


Extended Figure 3 | TF motif activity profiling from ChromVar and cross-modality correlation.

a, Workflow schematic for ChromVar analysis across four regulatory contexts. **b,** Variability of motif activity across metacells, comparing results from assessing enhancer CREs or promoter CREs using either the ATAC or TSS signals. Variability scores (n = 879 motifs) are shown with bootstrap confidence intervals. Top 5 most variable TF motifs are highlighted. **c**, Scaled ChromVar variability score and correlation coefficients for the 420 TF motifs with expression support across the four regulatory contexts. Motifs passing the thresholds are indicated in solid colors. **d**, Heatmap showing summarized motif activity across the 10 cell clusters with four conditions from 257 TF motifs. Values represent the average scaled ChromVar z-scores per clusters, illustrating dynamic regulatory transitions. **e**, Odd ratios from the enrichment between different modes of action identified in this study with established literature using different cutoffs from ChromVar variability (> 25% quantile, > median and > 75% quantile, with p.adj < 0.01) and from auto-correlation (>= 0.1 to 0.8). **f**, Promoter-types responded to the four TF regulatory modes. **g**, Upset plot summarizing the number of TF motifs categorized by their regulatory modes (n = 175 from promoter-enhancer joint ChromVar analysis). **h**, Percentage of TF motifs that are modality-specific according to the definition from **Fig. 3f** with different combinations of cutoff from ChromVar variability and correlation. **i**, Odd ratios from the enrichment between different regulatory modes identified with promoter-enhancer joint ChromVar analysis and established literature across 122 TF motifs with information from both ends.